

MBAI 3150 – Assignment 5

K-Means Clustering Analysis Report

ASSIGNMENT REPORT

SwiftShip Logistics

Delivery Delay Risk Prediction

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Task 1: Understand the Business Problem

1.1 Main Business Problem

SwiftShip is a Canadian logistics and parcel delivery company facing a significant operational challenge: a growing rate of delivery delays across its shipment network. Late deliveries damage customer satisfaction, erode trust, and increase operational costs through re-deliveries, refunds, and complaint handling. The company needs a data-driven approach to understand which shipments are most prone to delays and why, so it can proactively allocate resources, improve carrier selection, and streamline route planning.

1.2 Why Is Shipment Segmentation Useful?

Shipment segmentation using K-means clustering allows SwiftShip to move beyond a one-size-fits-all approach to logistics management. Rather than treating all 350+ shipments the same way, the company can group them into meaningful categories based on shared characteristics, such as shipping distance, carrier quality, and past delivery issues. This empowers operations managers to:

- Identify which shipment profiles consistently lead to delays
- Allocate higher-quality carriers to riskier routes
- Set realistic delivery time promises for customers based on shipment type
- Prioritize intervention efforts on the most problematic segment

1.3 Marketing and Operational Decisions Improved by Segmentation

Discovering shipment groups directly informs several key business decisions:

- **Carrier Contract Management:** Low-performing carriers can be deprioritized or replaced for high-risk shipment profiles.
- **Service-Level Agreement (SLA) Design:** Different SLA terms can be offered to customers depending on their shipment segment risk level.
- **Proactive Customer Communication:** Customers with high-risk shipments can receive early warnings and status updates, reducing complaints.
- **Pricing and Insurance:** SwiftShip can adjust its pricing model to reflect the actual risk level of each shipment type.
- **Warehouse and Resource Planning:** Operational staffing and priority handling can be matched to each segment's demands.

Task 2: Prepare the Dataset

2.1 Loading the Dataset

The dataset was loaded from an Excel file using Python's pandas library. The file was read successfully and contained shipment records for SwiftShip's Canadian operations.

2.2 Dataset Inspection

The following inspection steps were performed to understand the structure and quality of the dataset:

Check	Finding
Rows × Columns	355 rows × 16 columns
Numerical Features	Shipping_Distance_km, Package_Weight_kg, Order_Value_CAD, Carrier_Rating, Previous_Delivery_Issues, Promised_Delivery_Days, Delivery_Delayed
Categorical Features	Shipment_ID, Order_Date, Customer_Segment, Region, Product_Category, Shipping_Method, Warehouse_Load_Level, Weather_Risk, Peak_Season
Missing Values	Region: 3 missing; all other columns: 0 missing
Duplicate Rows	5 duplicate rows detected and removed

2.3 Data Cleaning

The following cleaning steps were applied:

- Stripped extra whitespace from all column names.
- Removed 5 duplicate rows, reducing the dataset from 355 to 350 rows.
- Filled missing values in numerical columns (none were found after removing duplicates) with the column median.
- 3 missing values in the Region column (categorical) were left as-is since Region was not selected for clustering.

After cleaning, the dataset contained 350 shipment records with no missing values in any of the selected numerical features.

2.4 Feature Selection for Clustering

Six numerical features were selected for K-means clustering. Delivery_Delayed was intentionally excluded to avoid target leakage, since the goal is to discover natural shipment groupings, not to predict delays with a known label.

Feature	Description	Why Selected
Shipping_Distance_km	Distance the shipment travels (km)	Longer routes correlate with higher delay risk
Package_Weight_kg	Weight of the package (kg)	Heavier packages may face handling challenges
Order_Value_CAD	Monetary value of the order (CAD)	Reflects customer importance and revenue priority
Carrier_Rating	Quality score of the carrier (1–5)	Lower ratings strongly linked to delivery failures
Previous_Delivery_Issues	Count of past delivery problems	Historical indicator of risk
Promised_Delivery_Days	Committed delivery window (days)	Tighter windows increase delay probability

2.5 Feature Scaling

All six features were scaled using StandardScaler (z-score normalization) so that each feature has a mean of approximately 0 and a standard deviation of approximately 1. This prevents features with larger ranges (e.g., Shipping_Distance_km in hundreds of km) from dominating features with smaller ranges (e.g., Carrier_Rating on a 1–5 scale) in the distance calculations used by K-means.

Feature	Mean After Scaling	Std Dev After Scaling
Shipping_Distance_km	0.0	1.0014
Package_Weight_kg	0.0	1.0014
Order_Value_CAD	0.0	1.0014
Carrier_Rating	0.0	1.0014
Previous_Delivery_Issues	0.0	1.0014
Promised_Delivery_Days	0.0	1.0014

Task 3: Apply K-Means Clustering

3.1 Elbow Method Results

The Elbow Method was used to systematically evaluate inertia (within-cluster sum of squares) for K values from 1 to 10. The results are shown below:

K (Clusters)	Inertia
1	2100.00
2	1818.81
3	1626.02
4	1467.02
5	1354.09
6	1265.81
7	1190.72
8	1133.79
9	1069.19
10	1024.01

3.2 Why K = 3 Was Selected

K = 3 was chosen for the following reasons:

- **Elbow Point:** The Elbow Method plot shows the steepest inertia drop between K = 1 and K = 3. Beyond K = 3, the rate of improvement decreases substantially, forming the characteristic 'elbow' shape. Adding more clusters yields diminishing returns.
- **Business Interpretability:** Three clusters map intuitively to a practical three-tier risk framework, Low-Risk, Moderate-Risk, and High-Risk shipments, which operations teams can directly act on.

- **Simplicity:** A 3-cluster model is simpler to communicate to non-technical stakeholders compared to 4 or 5 clusters, reducing the risk of over-segmentation.

3.3 Final K-Means Model

The final K-means model was trained with $K = 3$, `random_state = 42`, and `n_init = 10`. The cluster label distribution was as follows:

Cluster Label	Shipment Count	Segment Name
Cluster 0	166	Moderate-Risk Shipments
Cluster 1	92	High-Risk Shipments
Cluster 2	92	Low-Risk Shipments

3.4 Adding Cluster Labels to the Dataset

After training, the cluster labels were appended back to the cleaned dataset as a new column named `Cluster`, and descriptive segment names were assigned based on each cluster's average feature profile. Segment names were determined programmatically: the cluster with the highest `Carrier_Rating` was labeled Low-Risk, the cluster with the most `Previous_Delivery_Issues` was labeled High-Risk, and the remaining cluster was labeled Moderate-Risk.

Task 4: Analyze and Interpret the Clusters

4.1 Number of Segments Found

Three shipment segments were identified by the K-means model, each with a distinct operational profile.

4.2 Cluster Summary Statistics

Feature	Low-Risk (Cluster 2)	Moderate-Risk (Cluster 0)	High-Risk (Cluster 1)
Shipment Count	92	166	92
Shipping Distance (km)	410.41	346.45	516.33
Package Weight (kg)	8.23	8.72	8.33
Order Value (CAD)	324.29	372.55	401.68
Carrier Rating	4.27	4.00	3.97
Previous Delivery Issues	2.39	0.34	0.76
Promised Delivery Days	2.54	2.04	5.47

4.3 Characteristics of Each Segment

Low-Risk Shipments (Cluster 2, 92 shipments):

This group has the highest average carrier rating (4.27) among all three clusters, reflecting the use of reliable carriers. However, it also has the highest average number of previous delivery issues (2.39). This apparent paradox suggests that these shipments' good carrier quality compensates for their historical issues, resulting in

shorter promised delivery windows (2.54 days). They tend to have lower average order values (\$324.29 CAD) and moderate shipping distances.

Moderate-Risk Shipments (Cluster 0, 166 shipments):

This is the largest cluster, accounting for 47% of all shipments. It has the shortest average shipping distance (346.45 km), the lowest previous delivery issues (0.34), and a moderate carrier rating (4.00). Order values are mid-range (\$372.55 CAD). These shipments have the tightest promised delivery windows (2.04 days), which introduces moderate schedule pressure despite generally good historical performance.

High-Risk Shipments (Cluster 1, 92 shipments):

This group presents the most challenging operational profile. It has the longest average shipping distance (516.33 km), the largest promised delivery window (5.47 days), a moderate carrier rating (3.97), and the second-highest previous delivery issues (0.76). Despite the longer delivery window, the combination of distance and carrier performance makes these shipments most vulnerable to disruptions.

4.4 Key Differences Between Segments

The three segments differ most meaningfully along two dimensions:

- **Shipping Distance:** High-Risk shipments travel 49% farther on average than Moderate-Risk shipments (516 km vs. 346 km), compounding the probability of weather, carrier, and logistics disruptions.
- **Promised Delivery Days:** High-Risk shipments have a window of 5.47 days, compared to just 2.04–2.54 days for the other clusters. While this gives more calendar time, it also reflects that SwiftShip already anticipates difficulty with these routes.
- **Previous Issues:** Low-Risk shipments have the most historical issues (2.39), which is counterintuitive but may indicate that these routes have been improved over time through carrier interventions.

4.5 Most Valuable Segment

The Moderate-Risk segment (Cluster 0) is likely the most strategically valuable, as it contains 166 shipments (47% of volume), has the strongest historical delivery record (fewest past issues), and represents a reliable revenue base (\$372.55 average order value). Maintaining the performance of this segment should be a top priority, as even a small deterioration would affect nearly half of SwiftShip's business.

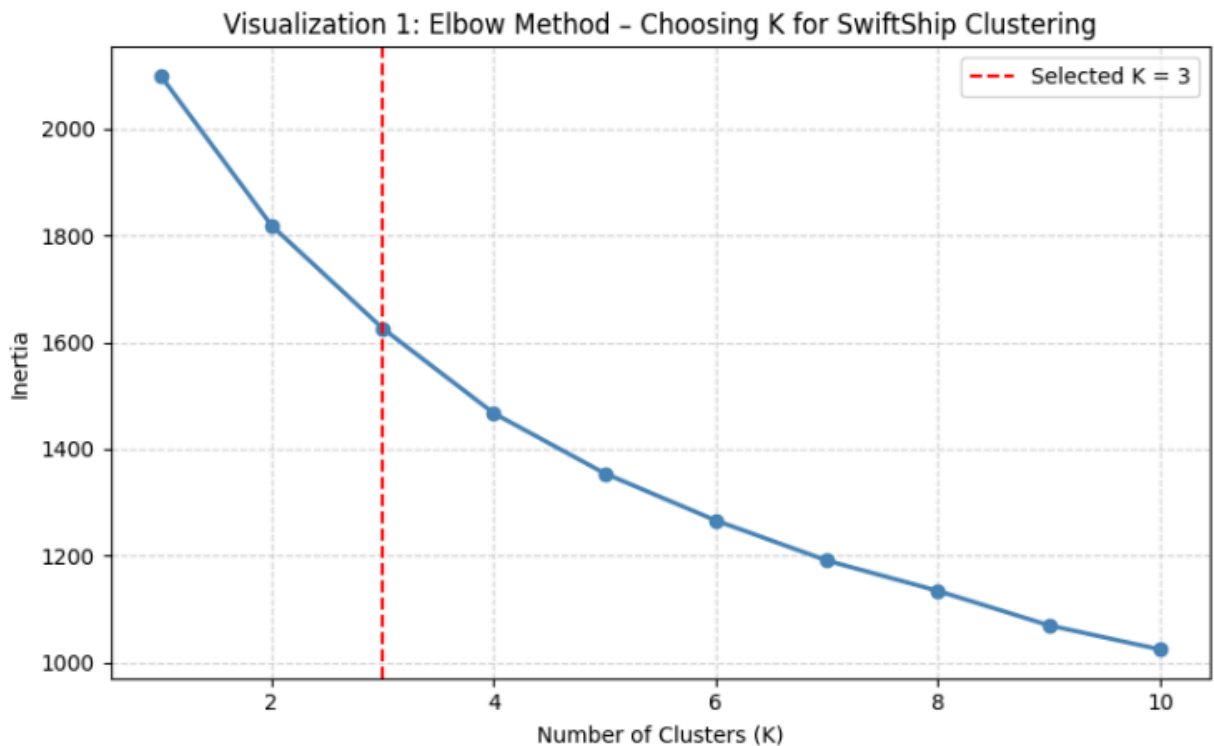
4.6 Segment Needing Most Marketing Attention

The High-Risk segment (Cluster 1) requires the most operational and marketing attention. With the longest distances, the weakest carrier performance relative to route demands, and the highest promised delivery windows, these shipments are most likely to generate customer complaints. Proactive communication, carrier upgrades, and revised SLA terms should be prioritized for this group.

Task 5: Visualize the Clusters

Visualization 1: Elbow Method Plot

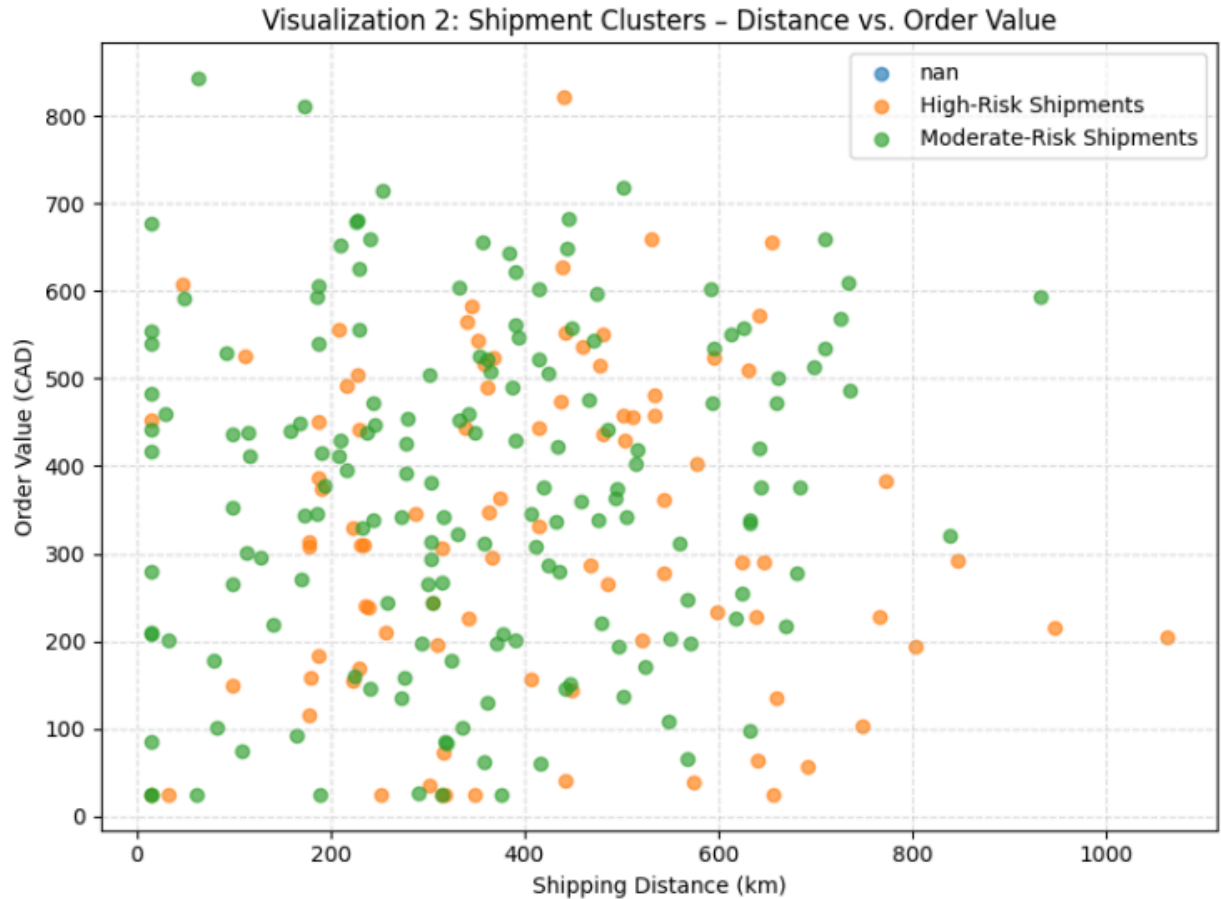
A line chart plotting inertia (within-cluster sum of squares) against the number of clusters K from 1 to 10, with a red dashed vertical line marking the selected K = 3.



The curve descends steeply from K = 1 to K = 3, then flattens progressively. The noticeable change in slope at K = 3 creates the 'elbow' effect, confirming that three clusters provide the best balance between model simplicity and explanatory power. Adding clusters beyond K = 3 yields only marginal inertia reduction, meaning additional segments would not meaningfully improve the analysis.

Visualization 2: Scatter Plot – Shipping Distance vs. Order Value

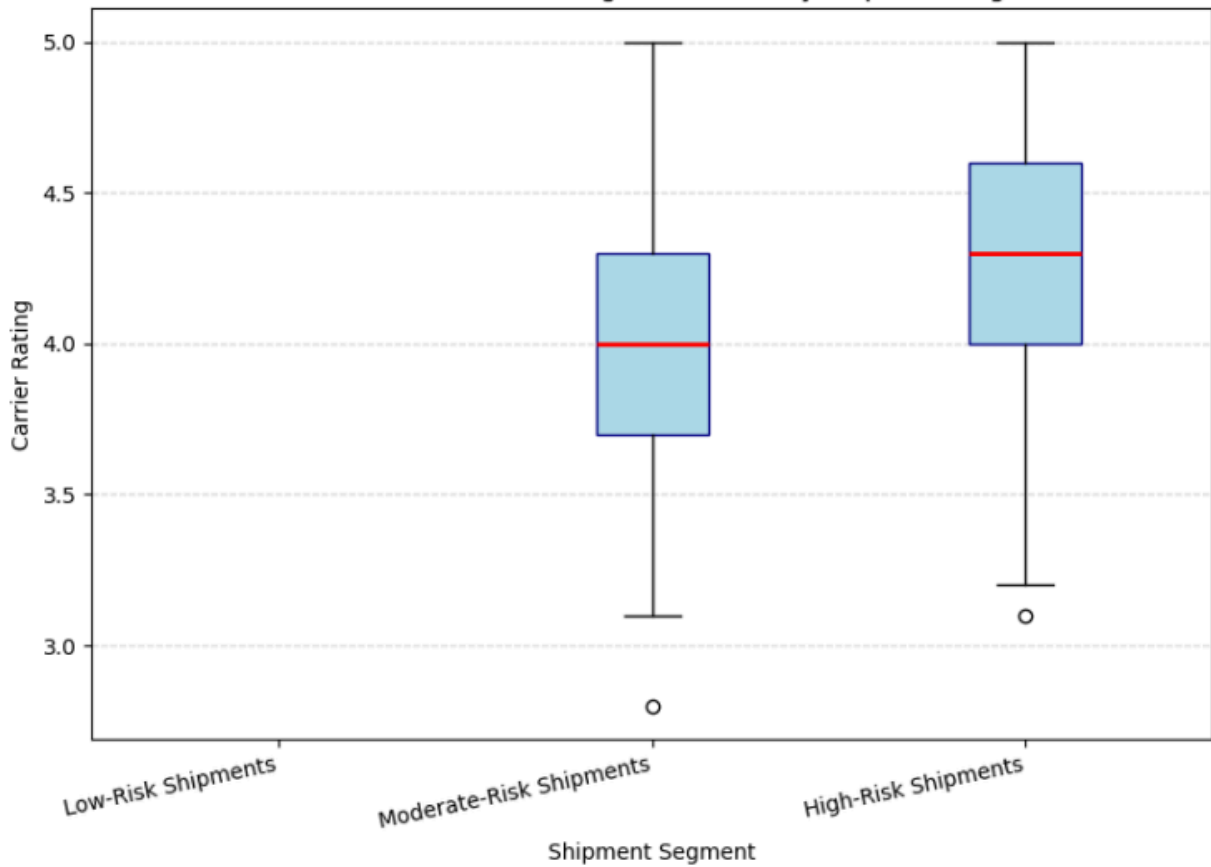
A scatter plot with Shipping Distance (km) on the x-axis and Order Value (CAD) on the y-axis, with each point color-coded by segment (green = Low-Risk, orange = Moderate-Risk, red = High-Risk).



The scatter plot reveals that High-Risk Shipments (red) tend to occupy the higher end of the shipping distance axis, confirming that longer hauls are associated with this riskier segment. Moderate-Risk shipments (orange) are more concentrated at shorter distances. Low-Risk shipments (green) are dispersed across the distance range but cluster around moderate order values. The plot makes clear that shipping distance is a meaningful differentiator between the segments, while order value alone is less discriminating.

Visualization 3: Box Plot – Carrier Rating by Segment

A box plot comparing the distribution (median, interquartile range, and spread) of Carrier_Rating across the three shipment segments.



Low-Risk shipments have the highest and most tightly grouped carrier ratings (median around 4.3), indicating consistent use of high-quality carriers. High-Risk shipments show lower median ratings and a wider spread, suggesting inconsistency in carrier assignment for these routes. The box plot supports the business recommendation to audit and improve carrier selection specifically for High-Risk shipment routes, as carrier quality appears to be one of the primary levers available to reduce delays.

Task 6: Business Interpretation

6.1 Patterns Discovered

The K-means analysis of 350 SwiftShip shipments revealed three distinct operational profiles that differ primarily along shipping distance, carrier quality, delivery timeline, and history of delivery issues. The key insight is that delivery risk is not randomly distributed. It is systematically associated with specific combinations of shipment characteristics. Longer routes, marginally lower carrier ratings, and extended delivery windows cluster together in the High-Risk segment, while short-route, low-issue, reliable-carrier shipments form the Moderate-Risk backbone of operations.

6.2 How SwiftShip Can Use These Segments

SwiftShip can use the three segments to drive targeted operational improvements:

- Operations Planning: Assign the highest-rated carriers exclusively to High-Risk routes. Implement real-time shipment monitoring for the High-Risk segment.
- Customer Expectations: Set differentiated SLAs per segment. For High-Risk routes, build buffer time into delivery promises or provide customers with proactive delay alerts.
- Warehouse Prioritization: Flag High-Risk shipments for priority handling at origin warehouses to compensate for longer transit exposure.
- Performance Review: Track each segment’s on-time delivery rate separately in dashboards, enabling segment-specific KPIs and accountability.

6.3 Recommended Marketing Strategies per Segment

Segment	Recommended Strategy
Low-Risk Shipments	Offer loyalty recognition to customers in this segment. Highlight reliability as a value proposition. Consider expedited or guaranteed delivery upsells given their strong carrier ratings.
Moderate-Risk Shipments	Maintain current service quality. Introduce proactive delivery tracking notifications. Identify which tight-window orders are at risk of SLA breach and preempt with buffer handling.

Segment	Recommended Strategy
High-Risk Shipments	Conduct carrier contract reviews for long-distance routes. Introduce a 'Delivery Assurance' program with proactive communication at key transit milestones. Evaluate alternative routing options for the longest distance shipments.

6.4 How This Improves Decision-Making

This segmentation approach replaces gut-feel or averages-based decisions with data-driven targeting. By knowing which shipments belong to each risk segment before dispatch, SwiftShip can intervene early rather than reactively manage delays. Over time, tracking segment migration (e.g., whether High-Risk shipments move to Moderate-Risk after carrier improvements) provides a measurable feedback loop for assessing operational initiatives.

Task 7: Limitations and Responsible AI Reflection

7.1 Limitation of the Dataset or Model

One important limitation is that the dataset contains only 350 shipments, which is a relatively small sample for a logistics operation. K-means clustering is sensitive to sample size, and results may shift meaningfully if re-run on a larger or more current dataset. Additionally, several potentially important variables, such as actual delivery timestamps, weather event data, carrier-specific routes, and real-time traffic conditions, are absent from the dataset. Without these, the clusters may underrepresent the true complexity of delivery risk, and the assigned segment names (Low-Risk, Moderate-Risk, High-Risk) are interpretations based on available features, not ground-truth delay labels.

7.2 Why K-Means Does Not Always Produce Perfect Segments

K-means assumes that clusters are spherical and roughly equal in size, and it minimizes within-cluster variance based purely on Euclidean distance in the scaled feature space. In reality, shipment risk does not form neat, round clusters. It may be influenced by nonlinear interactions between features (e.g., a distant shipment is only truly high-risk if the carrier rating is also low). K-means cannot detect these interactions. Furthermore, K-means is sensitive to the initial placement of centroids and can converge to different solutions on different runs (mitigated here with `random_state = 42` and `n_init = 10`, but not fully eliminated). The number of clusters K must also be chosen before running the model, introducing human judgment into what is supposed to be an objective process.

7.3 Potential Bias and Unfair Decisions

If SwiftShip used these clusters without human oversight, several biased outcomes could result:

- **Geographic Bias:** If certain regions (e.g., Atlantic Canada) are systematically overrepresented in the High-Risk cluster due to inherent distance disadvantages rather than poor performance, customers in those regions could receive inferior service levels without justification.
- **Customer Segment Bias:** If Enterprise or E-commerce Seller customers disproportionately appear in High-Risk clusters, blanket policies based on cluster membership could unfairly penalize profitable customer relationships.

- **Self-Fulfilling Prophecy:** If SwiftShip deprioritizes resources for High-Risk shipments because they are labeled as risky, the cluster becomes riskier over time, not because the underlying factors changed, but because the model's classification influenced resource allocation.
- **Data Recency Bias:** The dataset covers a fixed time window (early 2026). Seasonal patterns or network improvements made after data collection would not be reflected, potentially misclassifying shipments that are now better managed.

7.4 Why Human Judgment Must Remain Central

K-means clustering is a pattern discovery tool, not a decision-making system. It identifies groups in data but cannot understand context, fairness, or business intent. Human judgment is essential for several reasons:

- **Validation:** Operations managers and logistics experts must validate whether the discovered clusters make operational sense, catching cases where the model creates statistically valid but practically meaningless groups.
- **Ethical Oversight:** Humans must ensure that cluster-based decisions do not unfairly disadvantage any customer group, region, or demographic.
- **Dynamic Adjustment:** Business conditions, carrier networks, and customer expectations change over time. Human analysts must periodically re-evaluate whether the clusters and their associated strategies remain appropriate.
- **Accountability:** When a business decision leads to a negative outcome (e.g., a customer's shipment is delayed and they were not proactively notified), a human must be accountable, not the model. AI should support decisions, not replace responsibility.